Lab 4: Computer Literacy

1) **Number Conversions**
   These are meant to be done by hand. You may use a basic calculator if needed for math calculations, but you should be able to do these by using conversion techniques.

$_2$	a. $(356)_{10} = (101100100)_2$
$_{10}$	b. $(11010010)_2 = (210)_{10}$
$_{16}$	c. $(1\ 1110\ 1100\ 1101)_2 = (1ECD)_{16}$
$_2$	d. $(B35)_{16} = (101100110101)_2$
$_2$	e. $(572)_8 = (101111010)_2$
$_8$	f. $(11\ 111\ 110\ 010)_2 = (3762)_8$

2) Describe the hardware components of your personal computer or of the computer in our lab to which you have access**. Include the processor type and speed, storage capacities of main memory(RAM), and types of I/O devices.** Explain how you determined your answer.

CPU: 12th Gen Intel(R) Core(TM) i7-12700 (2.10 GHz)
Storage: 477 GB
RAM: 32 GB
I/O: Mouse/Keyboard

3) How many unique items can be represented with each of the following?

7	a. 3 bits
32	b. 5 bits
1,024	c. 10 bits

4) If a parking garage has 200 parking slots, what is the minimum number of bits needed to number each slot? Why?

They would need 8 bits to be able to store each character. Because without 8 bits they wouldn't have enough possibilities to store the characters.

5) If a language uses 240 unique letters and symbols, how many bits would be needed to store each character? Why?

They would need 8 bits to be able to store each character. Because without 8 bits they wouldn't have enough possibilities to store the characters.

6) How many bits are there in each of the following? How many bytes are there in each?

	bits	bytes
a. 9 KB	72000	9000
b. 15MB	1.2 x 10^8	1.5 x 10^7
c. 2 GB	1.6 x 10^10	2 x 10^9